| Enrolment Number | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|

*Total No. of printed pages = 04*

**Winter, 2026**

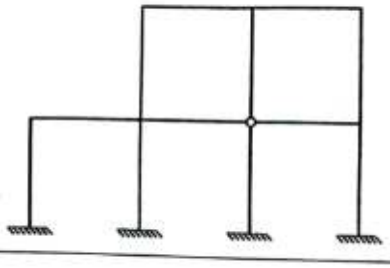
**B. Tech 4th Semester Examination**

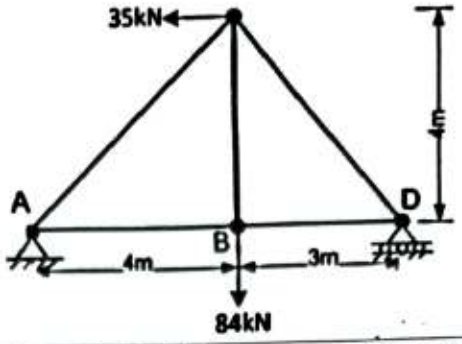
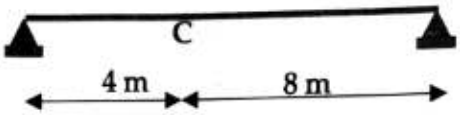
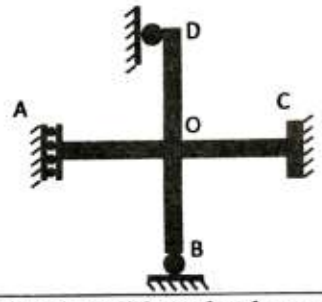
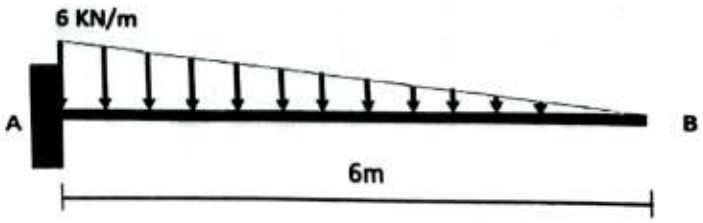
**Structural Analysis – Theory**

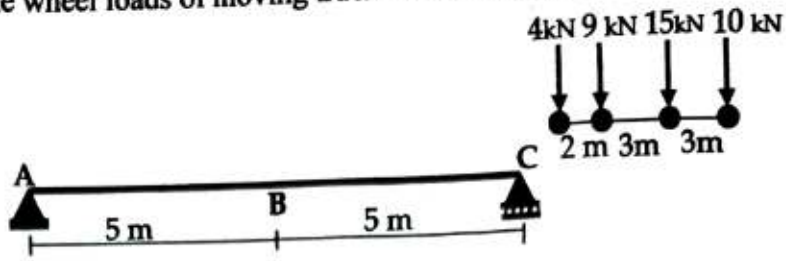
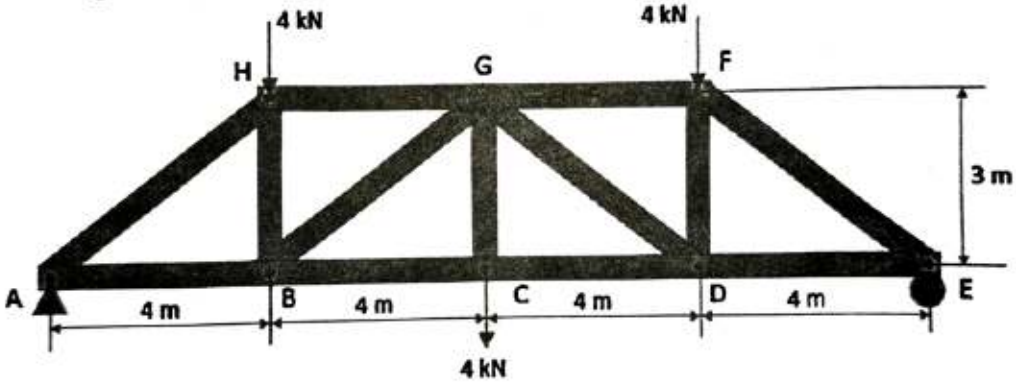
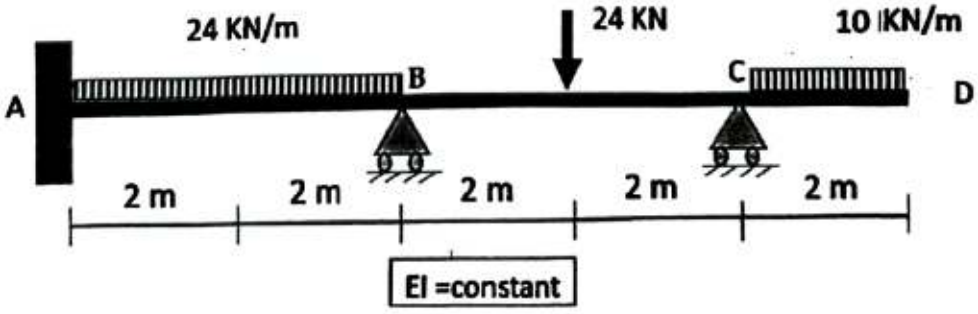
**Course Code: BCE23405T**

**Full Marks: 50** **Time: 2 hours**

*The figure in the margin indicates full marks for the questions.*

| S. No | Questions | Marks |
|---|---|---|
| | **Section 1 [10 Marks]** | |
| 1a | 1.If the degree of static indeterminacy of a structure is zero, the structure is called ..... | 1 |
| 1b | Influence lines are especially useful for structures subjected to .....loads. | 1 |
| 1c | What do you mean by stiffness of a member. | 1 |
| 1d | Define distribution factor. | 1 |
| 1e | Rankine's formula combines.....and Euler theory. | 1 |
| 1f | Explain principal of superposition. | 1 |
| 1g | The effective length of a cantilever column is ..... | 1 |
| 1h | Castigliano's theorem is applicable to structures behaving within the..... range. | 1 |
| 1i | The fixed end moment for a beam carrying a point load at the midspan is .....at both ends. | 1 |
| 1j | The method of joints is mainly used to determine..... | 1 |
| | **Section 2 [8 Marks]** Answer any 4 questions [2 marks each] | |
| 2a | Determine the degree of indeterminacy for the rigid-jointed frames indicated in Figures.  | 2 |

| | | |
|---|---|---|
| 2b | Classify the truss in the figure below as stable, unstable, statically determinate, or static indeterminate. Also calculate the reactions of the given truss.<br>  | 2 |
| 2c | A three hinged parabolic arch of uniform cross section of span 60m and rise 10m carries a uniformly distributed load of 10 kN/m over the entire span. Calculate the vertical reactions and the horizontal thrust at supports. | 2 |
| 2d | Draw the influence line diagram for moment at point C of the given beam<br>  | 2 |
| 2e | Calculate the Distribution factor for each member of the beam shown in figure below. EI and L is constant for all the members.<br>  | 2 |
| 2f | What is the ratio of the strength of a solid steel column of 200mm diameter to that of a hollow circular steel column of the same cross-sectional area and a wall thickness of 20mm. The two columns have the same length and same boundary condition. | 2 |
| | **Section 3 [12 Marks]**<br>Answer any 3 questions [4 marks each] | |
| 3a | Derive the expression for crippling load for columns with one end fixed and other hinged. | 4 |
| 3b | Determine the maximum deflection of the beam E = 200 $GPa$ and I=100x10$^6$ mm$^4$.Use unit load method.<br>  | 4 |

| | | |
|---|---|---|
| 3c | Determine the maximum positive shear created at point B in the beam shown in Figure due to the wheel loads of moving truck. Draw the influence line for shear.<br>  | 4 |
| 3d | In the analysis of portal frame ABCD with sway, calculate the Restoring force (R) at the top of the frame. The end column moments are as follows:<br>$M_{AB}$ = 9.58 kN-m $M_{CD}$ = -15.00 kN-m<br>$M_{BA}$ = 19.34 kN-m $M_{DC}$ = 0 kN-m<br>The length of column AB and CD are 10 m and 15 m respectively. | 4 |
| | **Section 4 [20 Marks]**<br>Answer any 2 questions [10 marks each] | |
| 4a | Calculate the forces in each member of the truss using **method of joints**. Support A is hinged and E is a roller support.<br>  | 10 |
| 4b | Analyse the beam shown in Figure using **Slope deflection method** if support B sinks by 20 mm. Draw the bending moment diagrams for the beam. Take EI = $2x10^4$ kN-$m^2$<br>  | 10 |